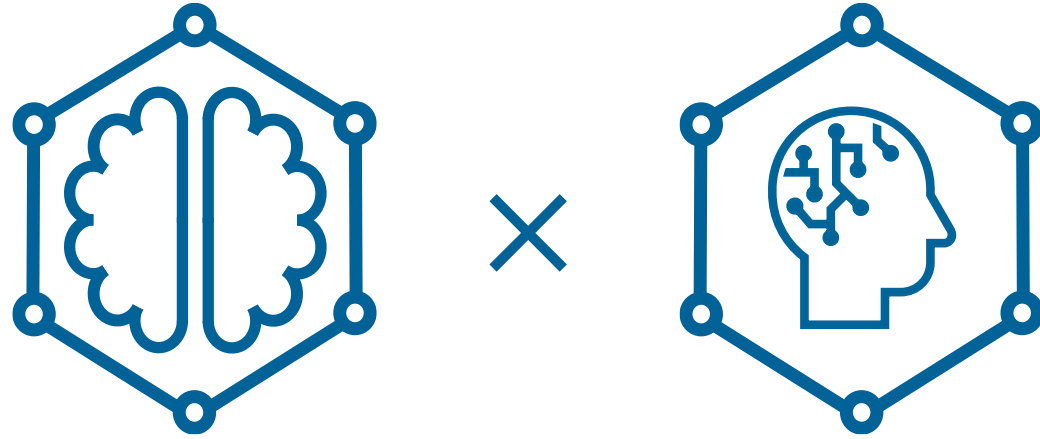




Knowledge Graphs

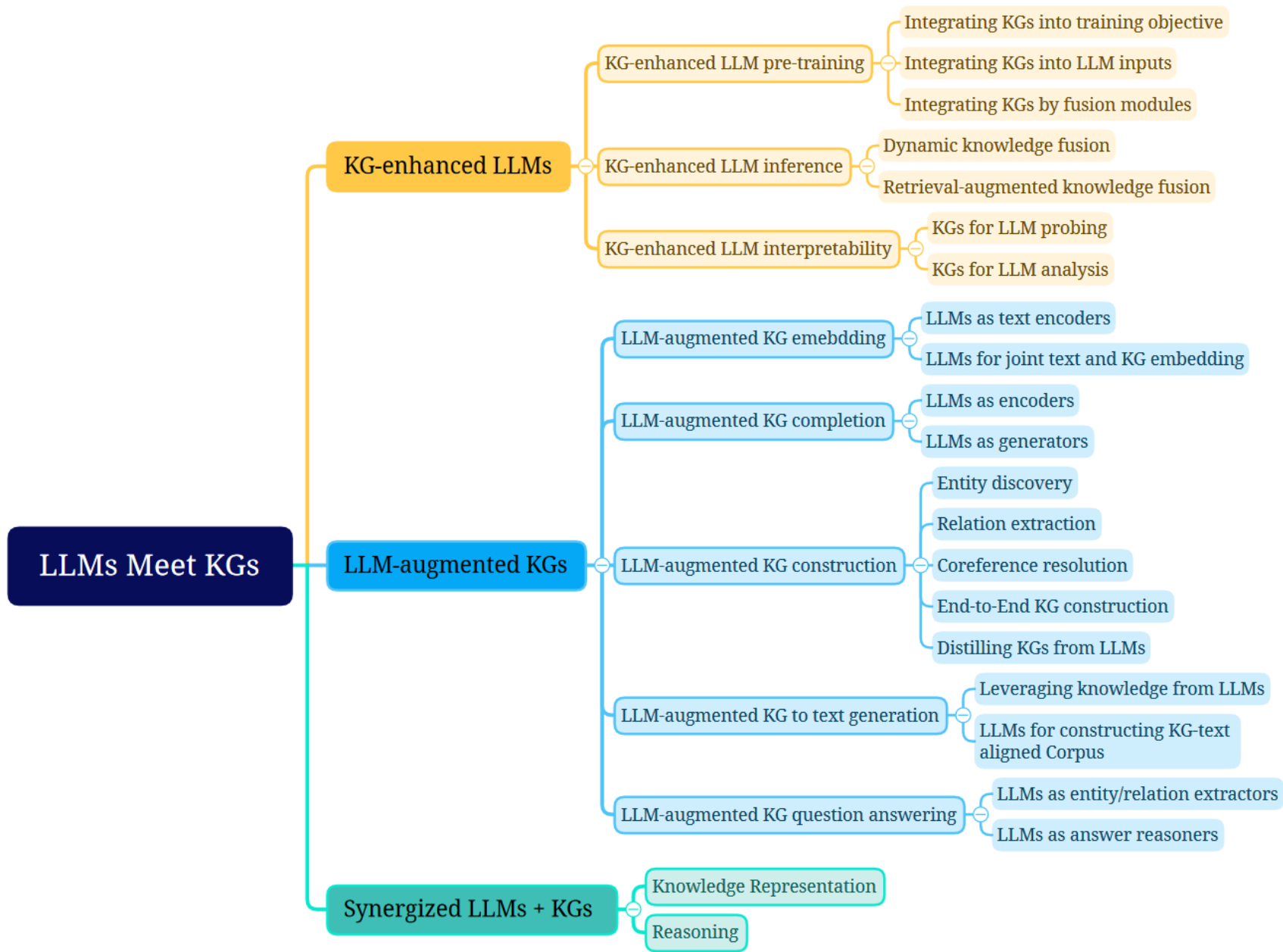


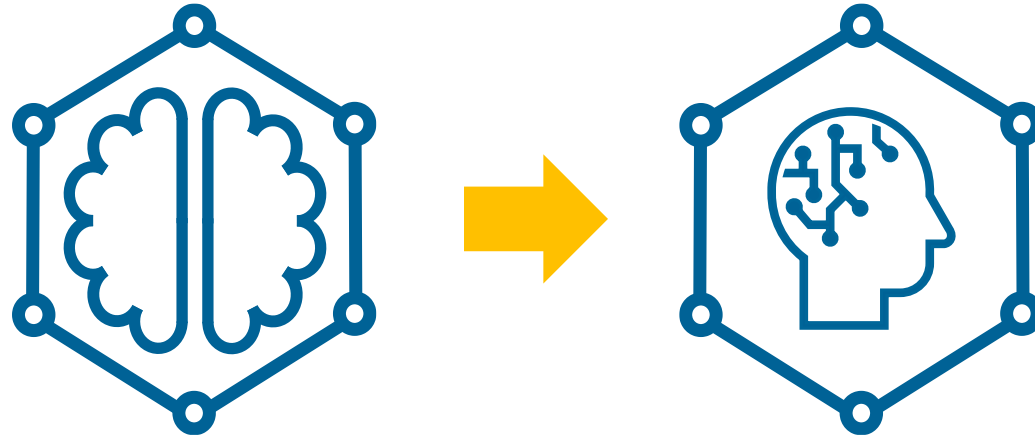
Generative AI



Knowledge Graphs × LLMs
KG x LLM Methods

Emanuel Sallinger



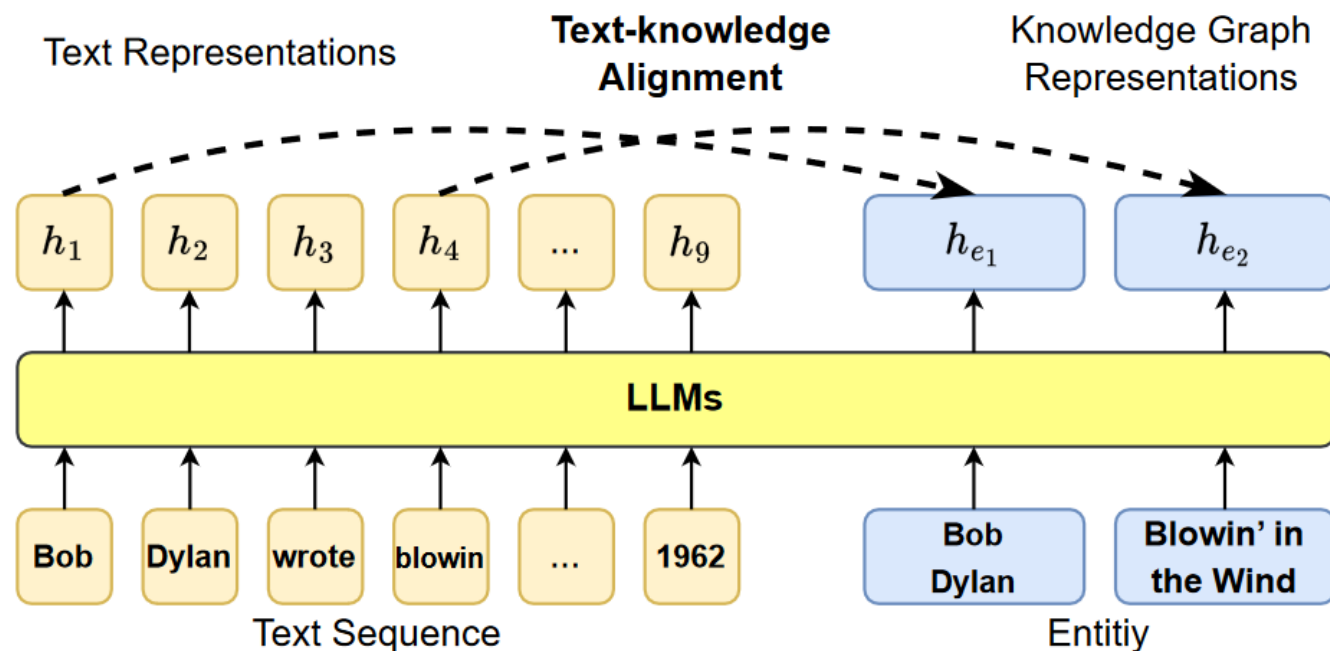


KGs For Enhancing LLMs



Enhancing Pre-Training

Injecting KG into LLM Training Objective

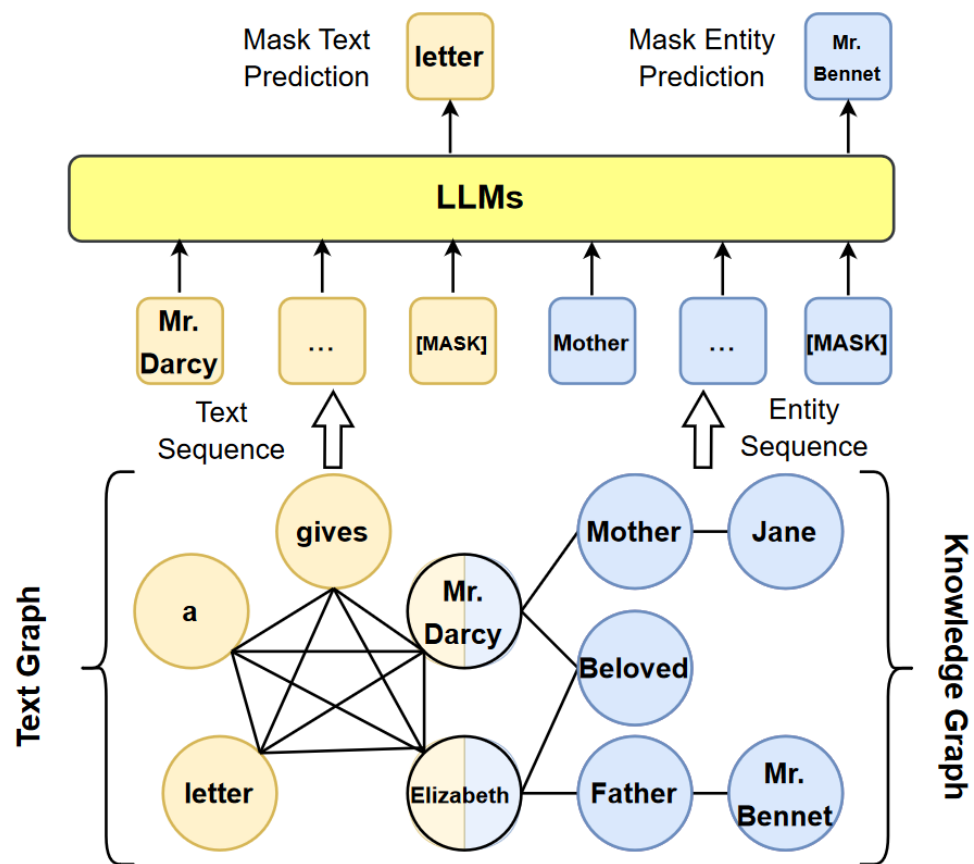


Input Text: Bob Dylan wrote Blowin' in the Wind in 1962



Enhancing Pre-Training

Injecting KG into LLM Input

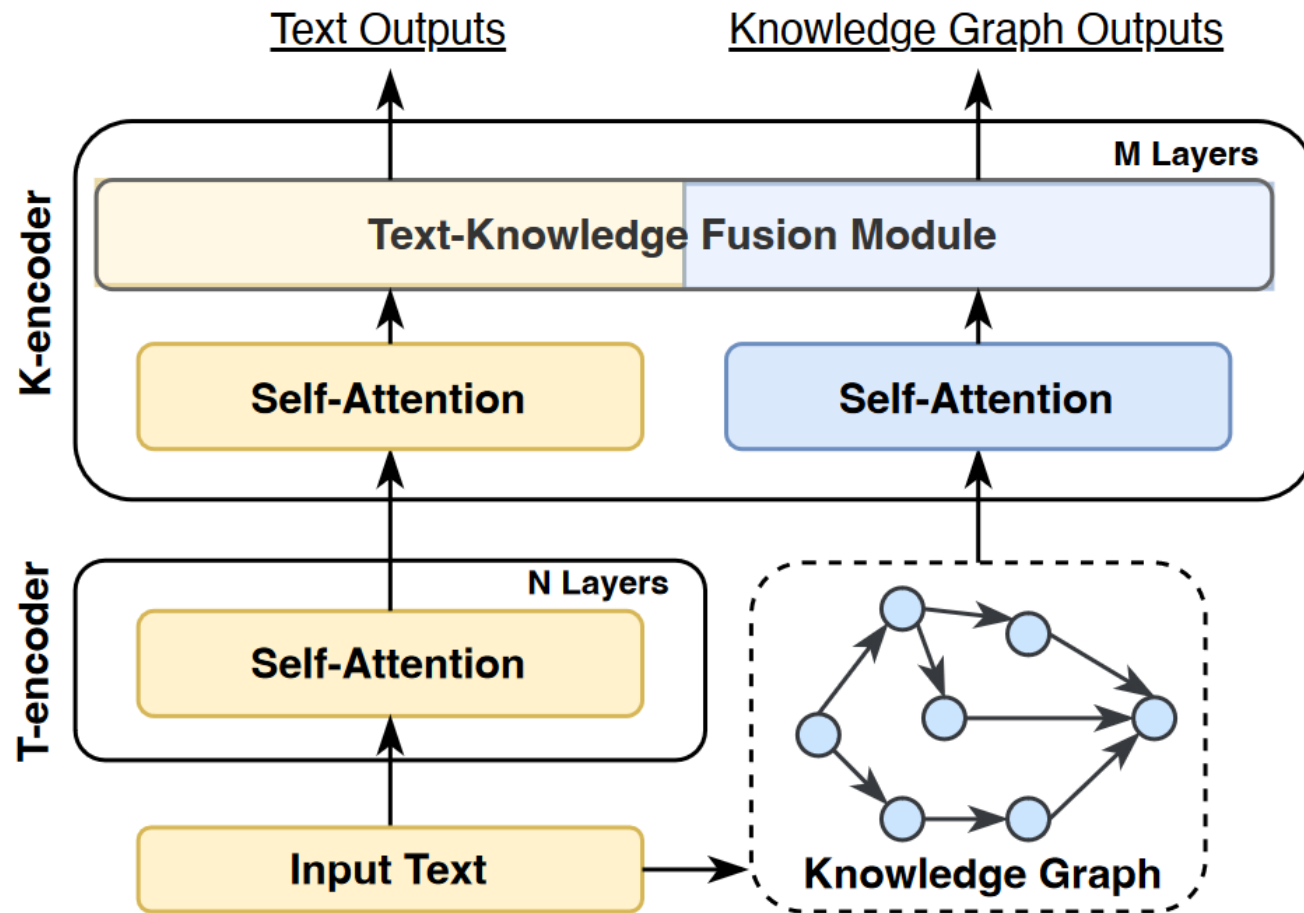


Input Text: Mr. Darcy gives Elizabeth a letter



Enhancing Pre-Training

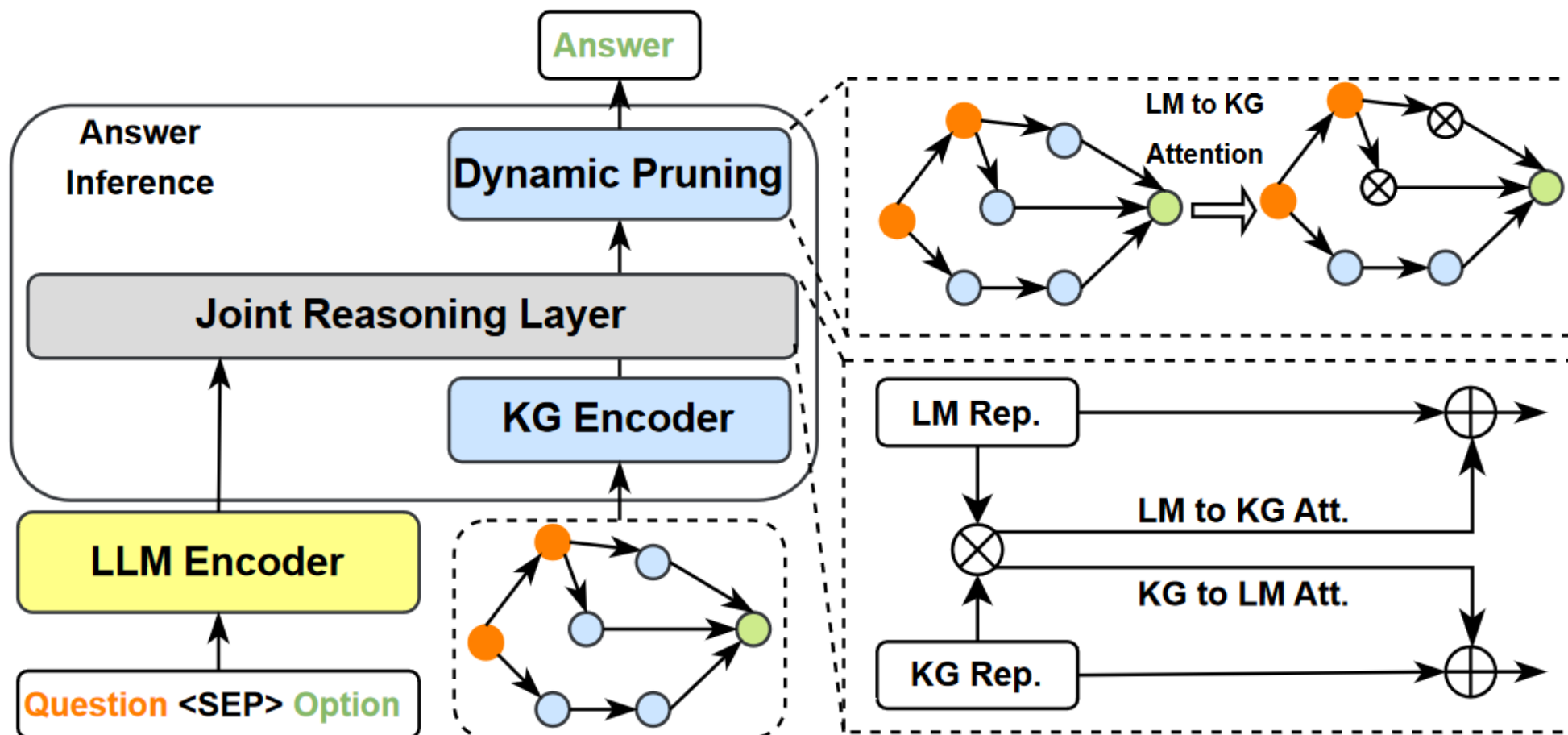
Fusion Module





Enhancing Inference

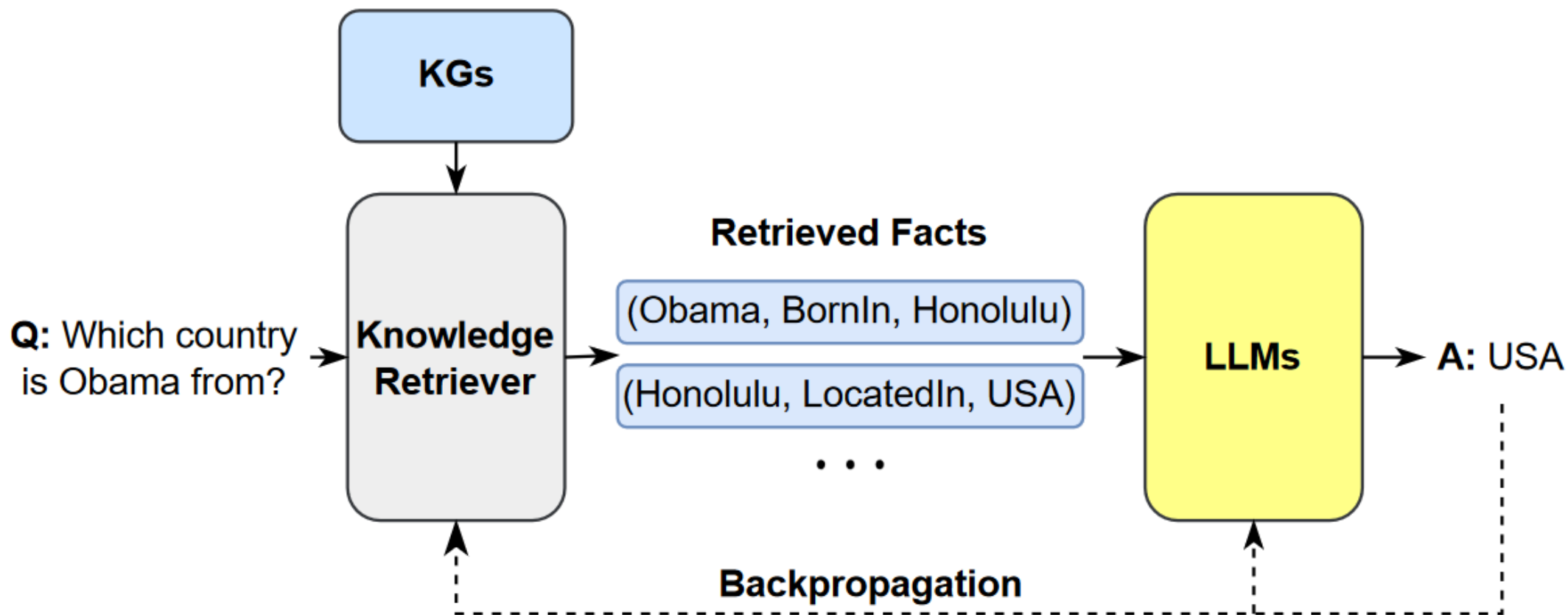
Dynamic Knowledge Fusion





Enhancing Inference

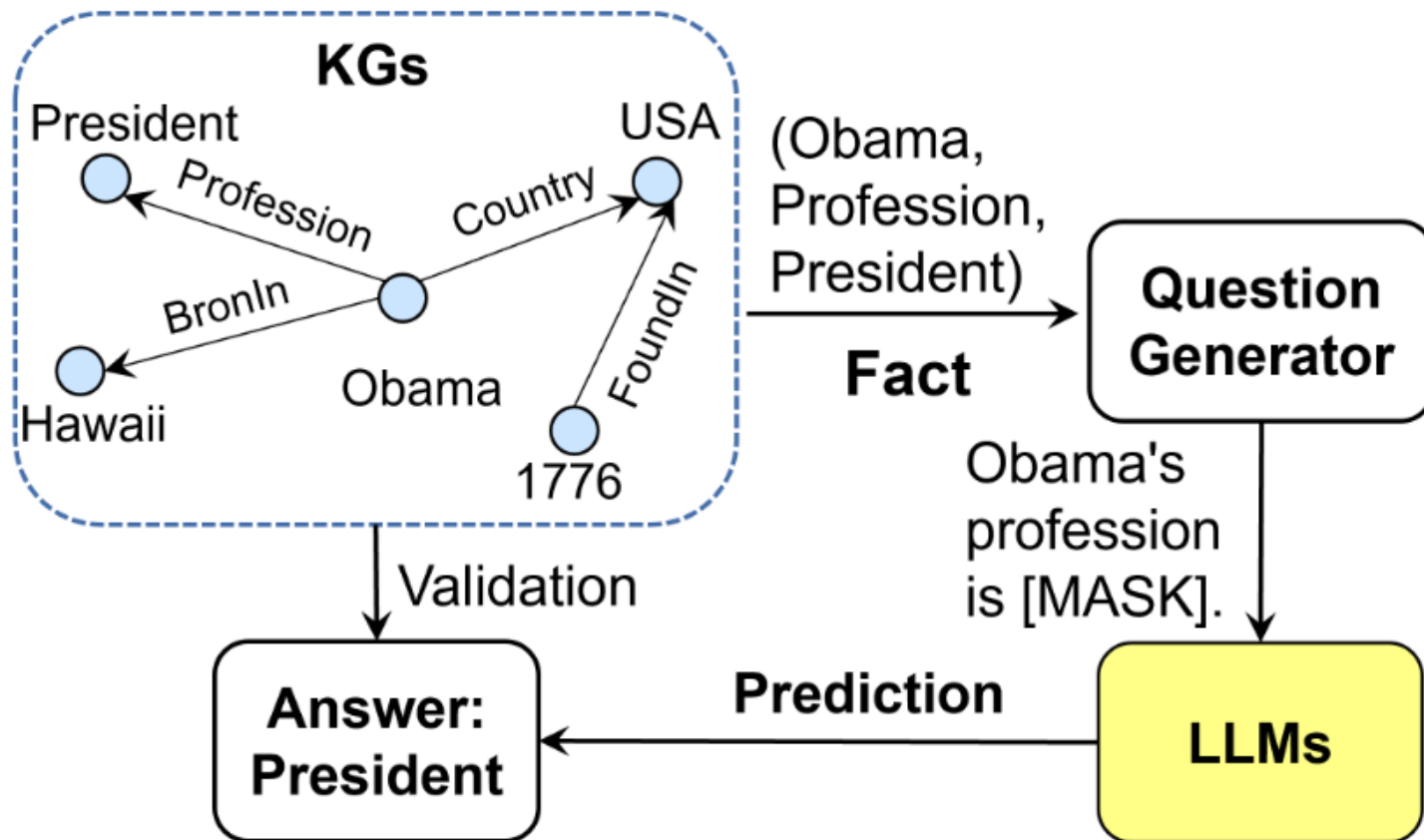
Retrieval-Augmented Knowledge Fusion





Enhancing Interpretability

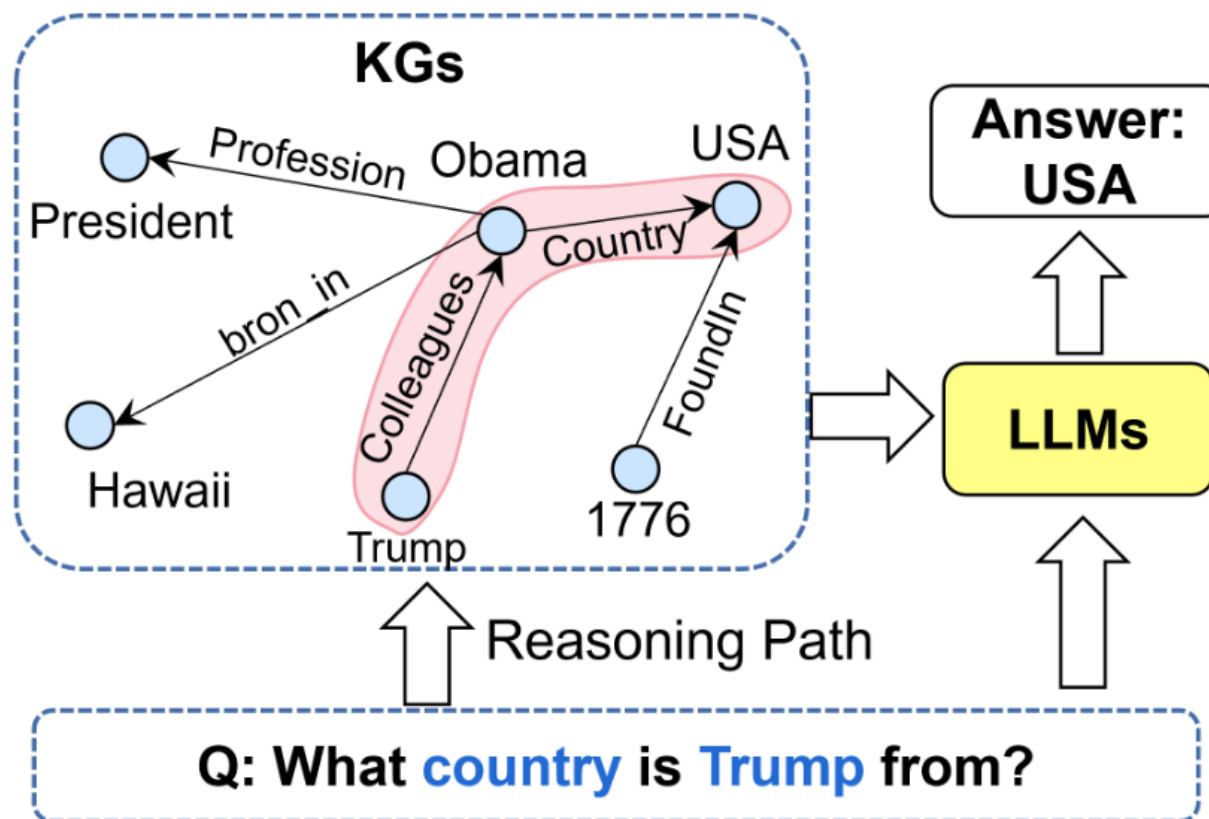
KGs for LLM Probing

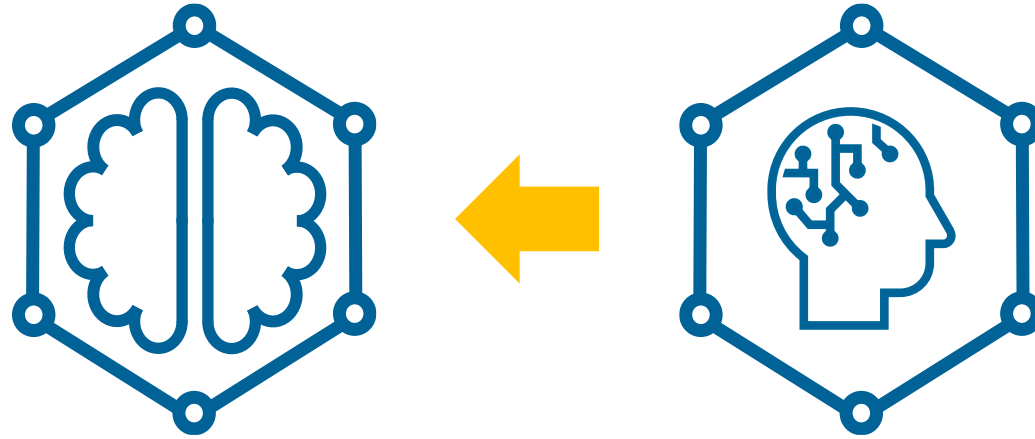




Enhancing Interpretability

KGs for LLM Analysis



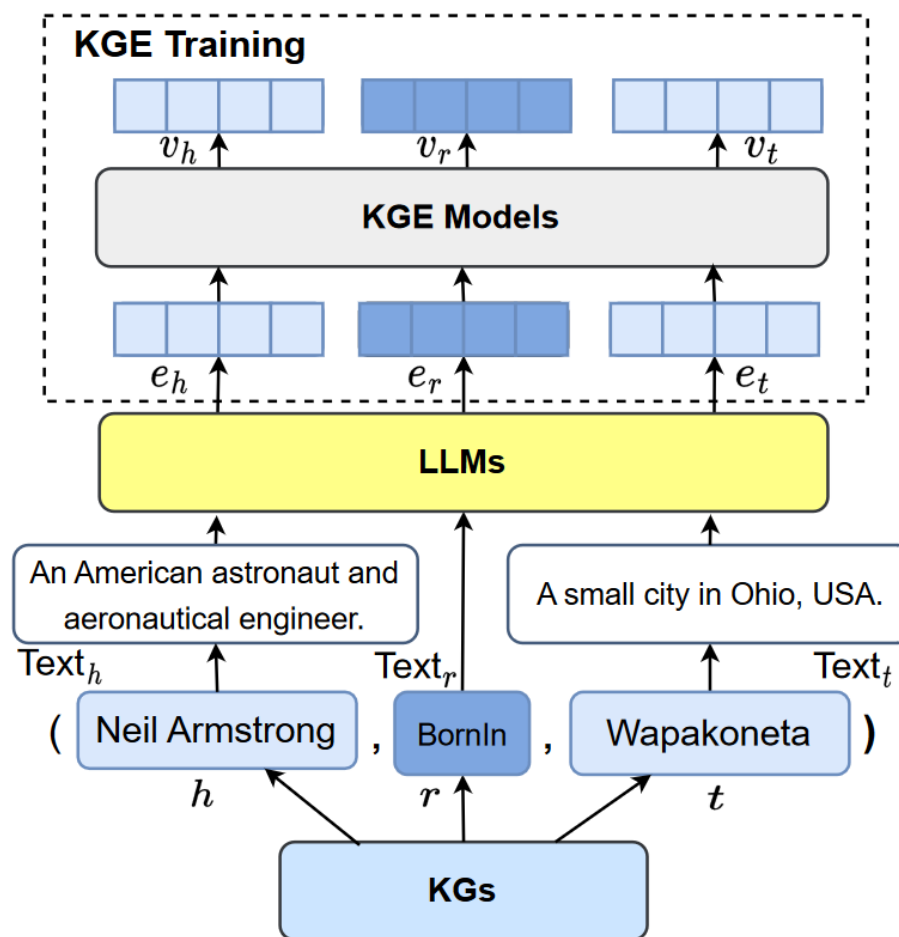


LLMs For Enhancing KGs



Enhanced KG Embeddings

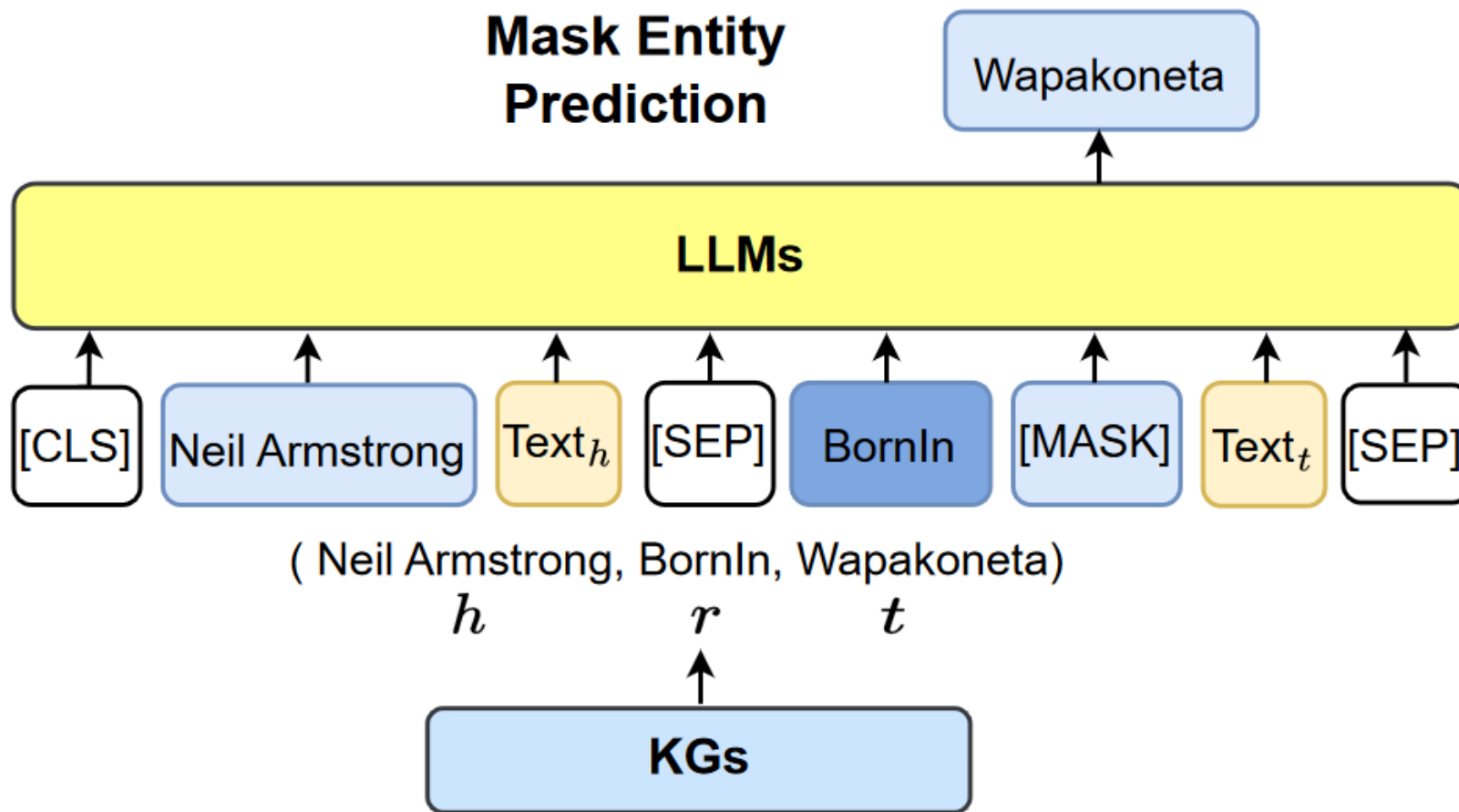
LLMs as Text Encoders





Enhanced KG Embeddings

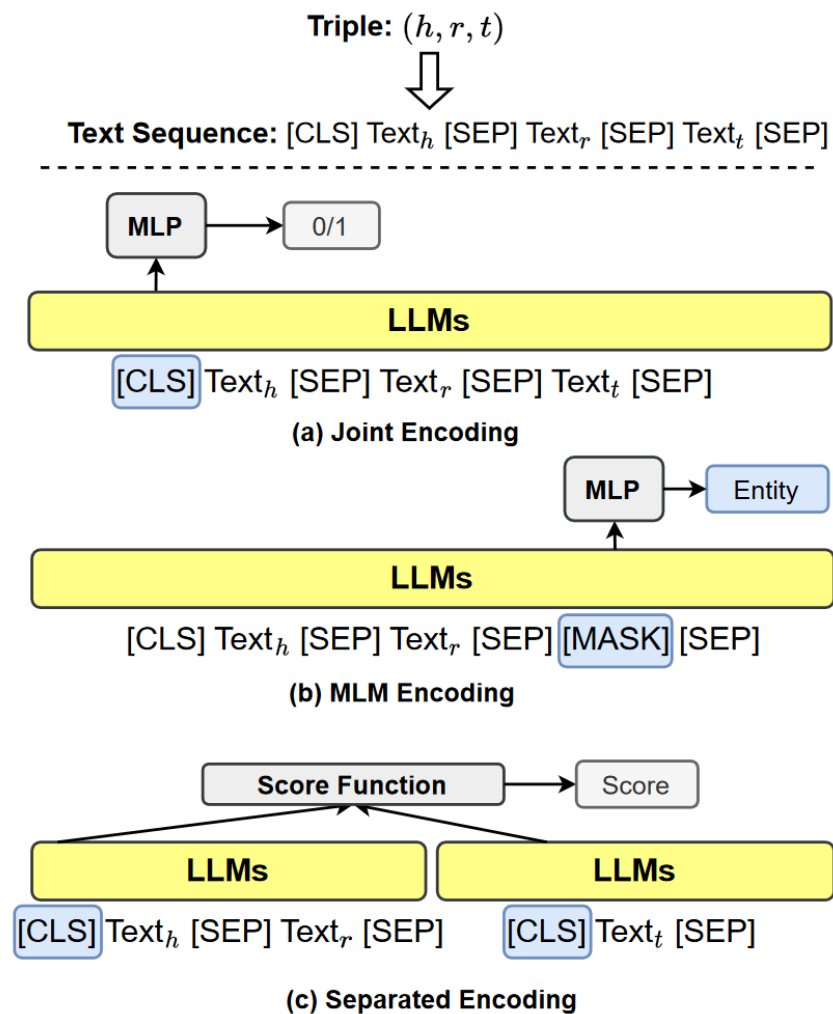
Joint Text and KG Embedding





Enhanced KG Completion

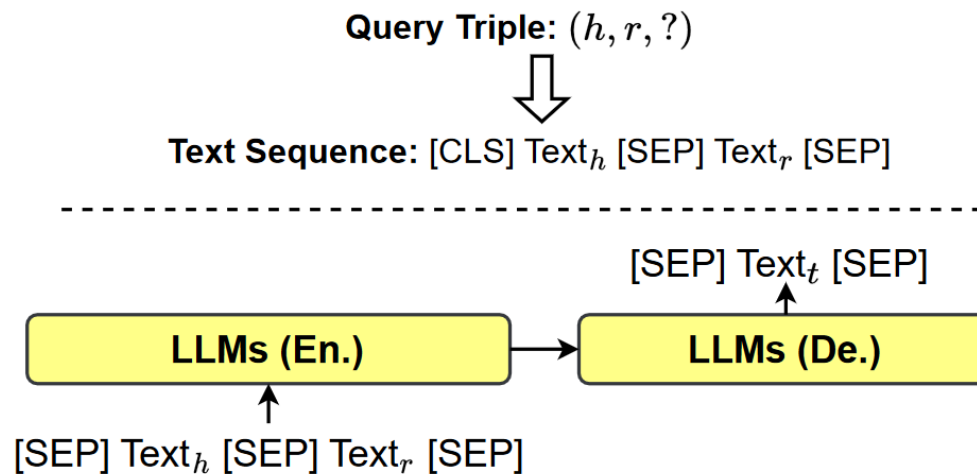
LLMs as Encoders



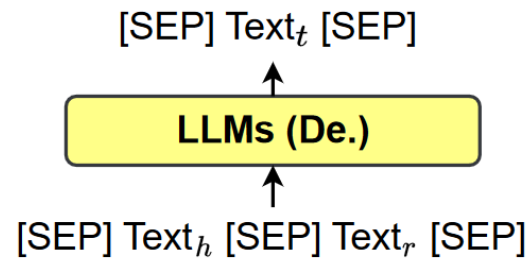


Enhanced KG Completion

LLMs as Generators



(a) Encoder-Decoder PaG

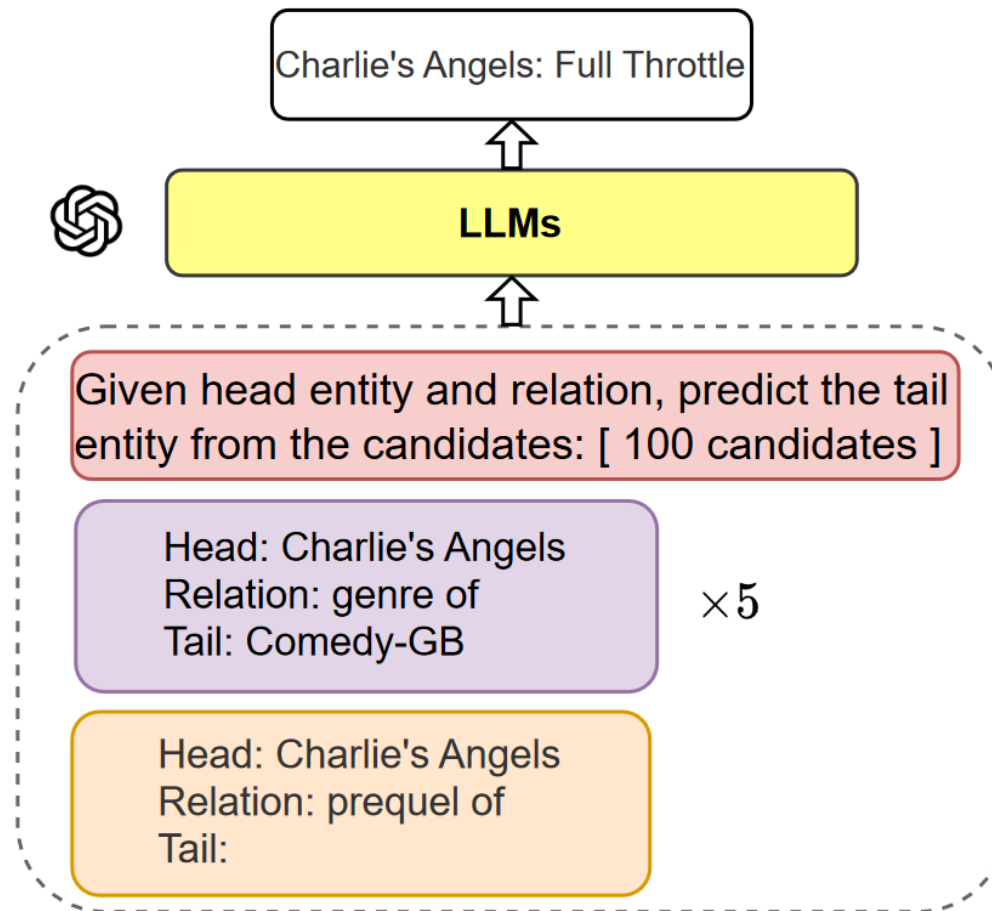


(a) Decoder-Only PaG



Enhanced KG Completion

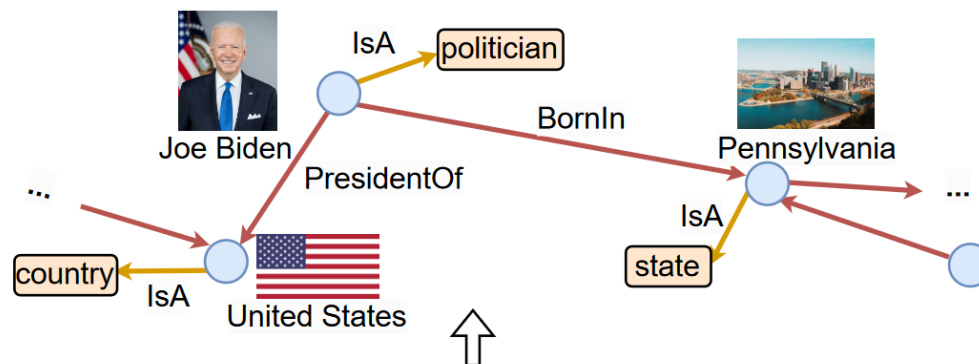
LLMs as Generators



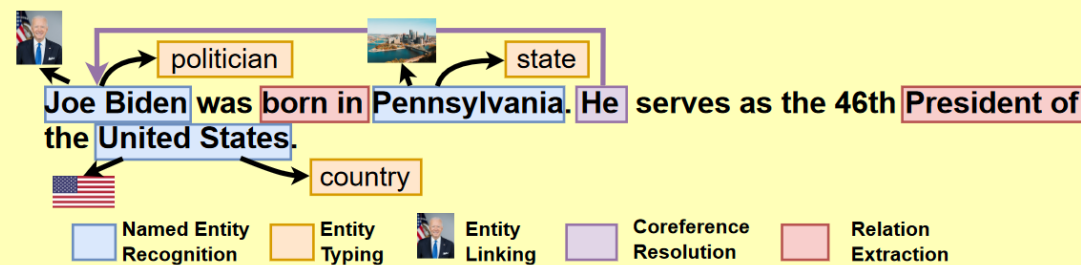


Enhanced KG Construction

Knowledge Graph



LLM-based Knowledge Graph Construction

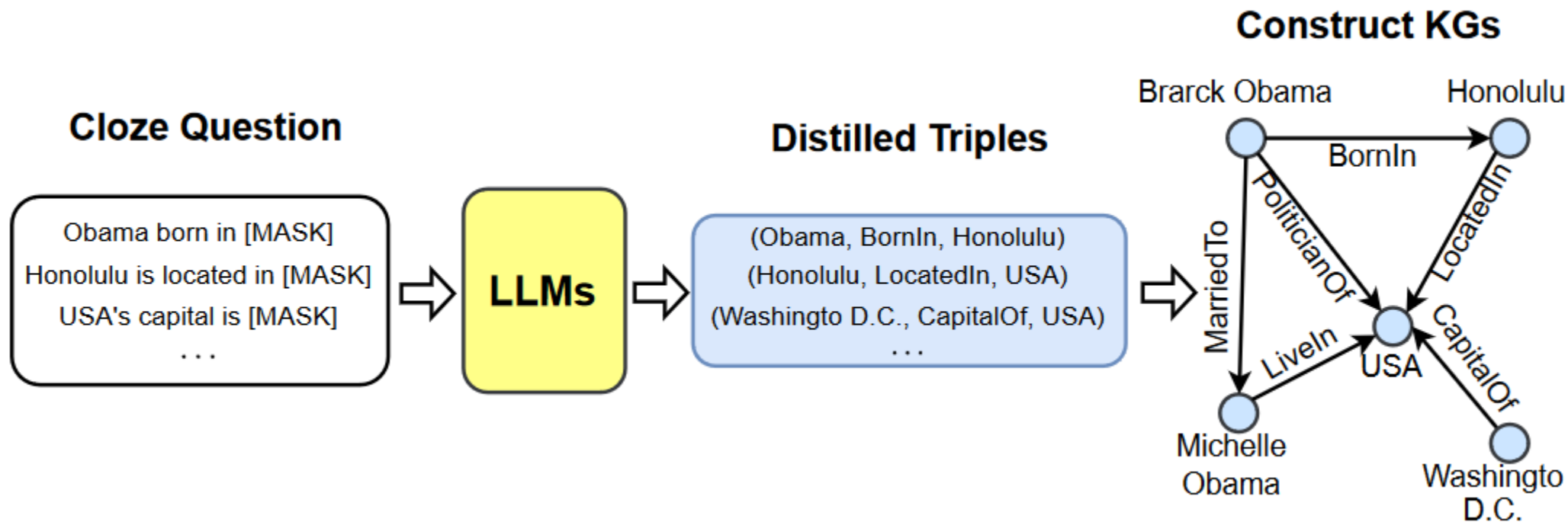


Text: Joe Biden was born in Pennsylvania. He serves as the 46th President of the United States.



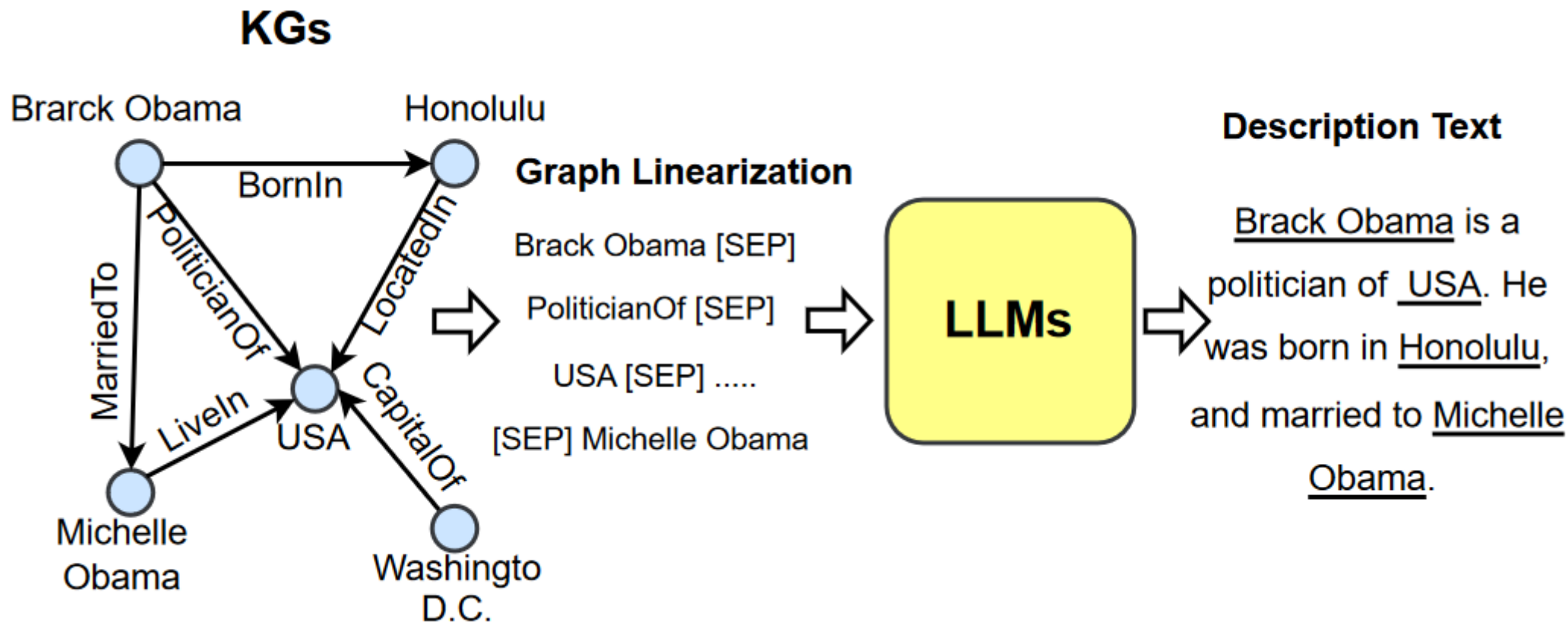
Enhanced KG Construction

Distilling KGs from LLMs





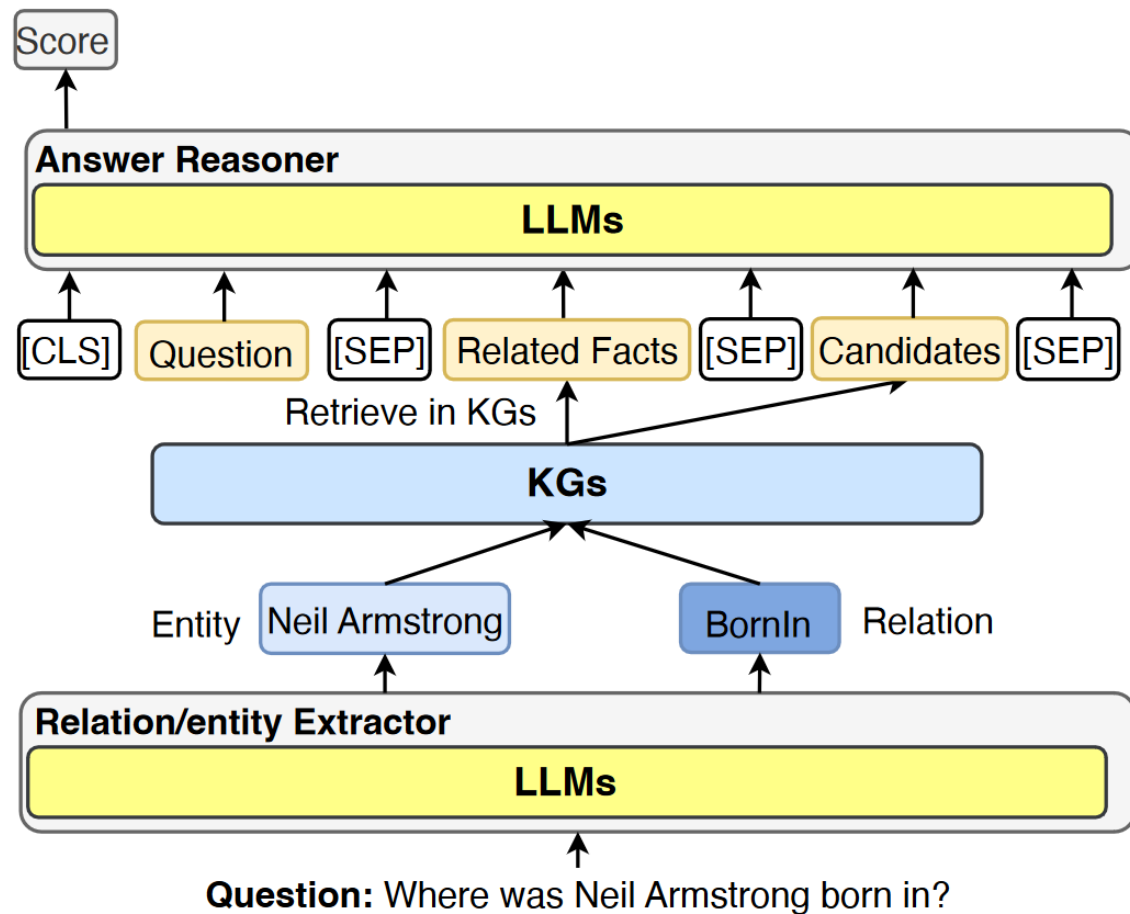
Enhanced KG to Text

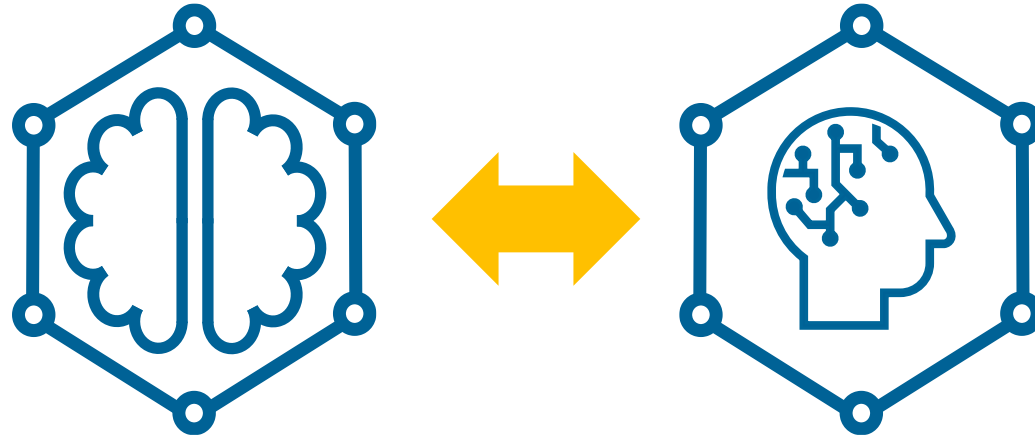




Enhanced Question Answering

LLMs as Entity/Relation Extractors and Answer Reasoners

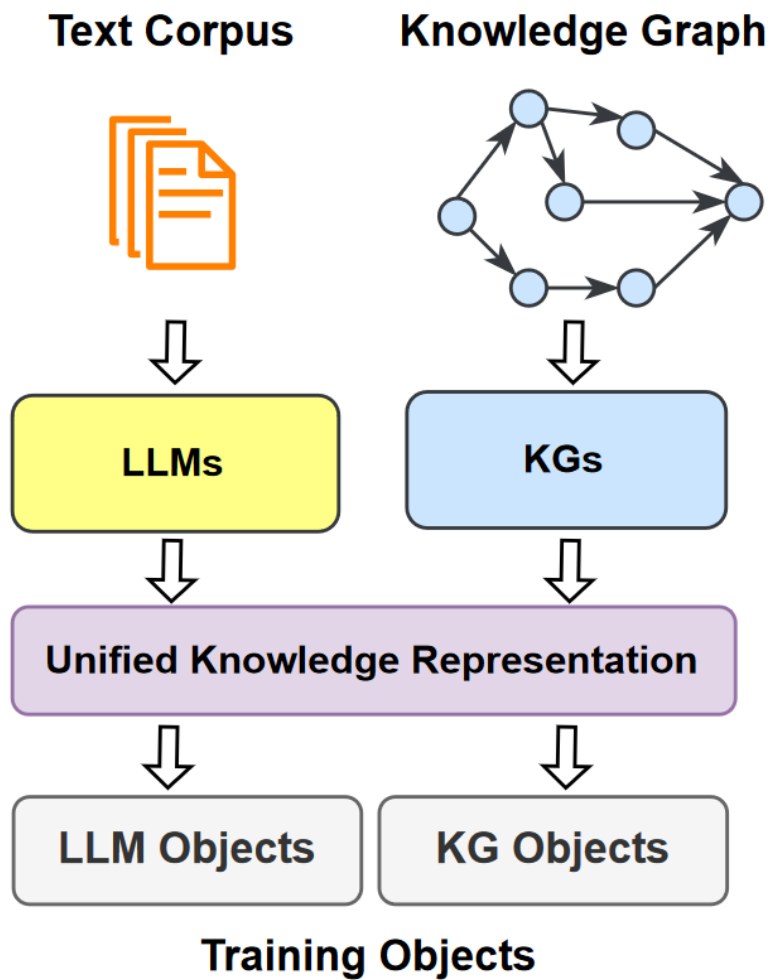




Synergetic KGs+LLMs



Knowledge Representation

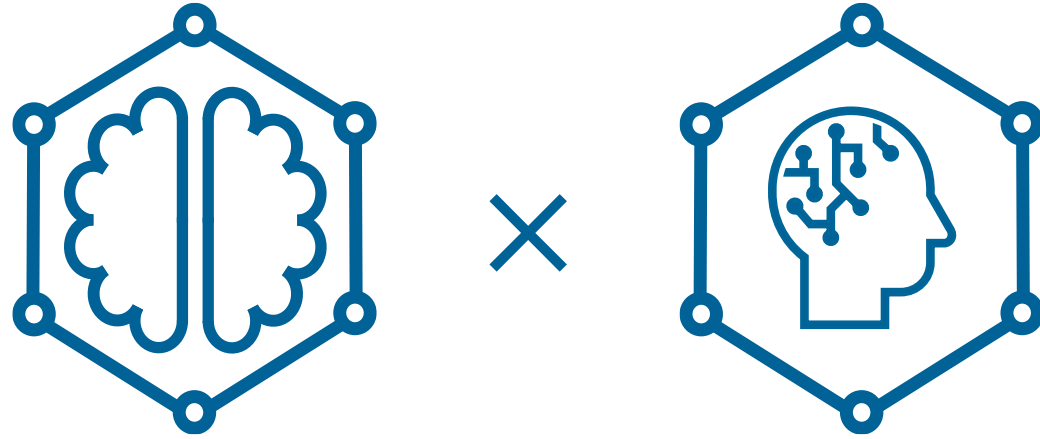




Knowledge Graphs



Generative AI



Knowledge Graphs × LLMs
KG x LLM Methods

Emanuel Sallinger